

Unit I

Introduction to Statistics

Learning Outcomes:

- Students should be able to understand the meaning of statistics.
- Students should be able to understand the importance of statistics.
- Students should be able to understand the scope of statistics.
- Students should be able to understand the concept of data and types of data.

Introduction

What is Statistics?

Statistics is the branch of science that deals with the collection, organization, presentation, analysis, interpretation, and drawing conclusions from data.

Statistics helps us make decisions based on data instead of assumptions.

“Statistics” is a word that has been derived from the Latin word ‘Status’ that implies a gathering of numbers or figures; those address some data of our human interest.

We discover Statistics in regular day to day existence, such as in books or other information papers or TV or newspapers. Although, in the first place it was utilized by Kings just for gathering data about states and other data which was required about their people, their number, income of the state etc. In India, it is felt that it began tracing all the way back to Chandra Gupta Maurya's realm under Chankya to gather the information of births and deaths. In any case, presently its extension has expanded. It is presently utilized in practically all fields of human information and abilities like Business, Commerce, Economics, Social Sciences, Politics, Planning, Medicine and different sciences, Physical just as natural.

Definitions of Statistics:

i) Webster Definition: The classified facts respecting the condition of the people in a state—especially those facts which can be stated in numbers or in tables of numbers or in any tabular or classified arrangement.

ii) Horace Sacrist Definition: By statistics we mean aggregates of facts affected to a marked extent by multiplicity of causes, numerically expressed, enumerated or estimated according to a reasonable standard of accuracy, collected in a systematic manner for a predetermined purpose, and placed in relation to each other.

iii) Croxton and Cowden Definition: Statistics may be defined as the collection, presentation, analysis, and interpretation of numerical data.

So, from the above definitions we find that the science of statistics also includes the methods of collecting, organizing, presenting, analysing and interpreting numerical facts and decisions are taken on their basis.

1.2 Importance of Statistics

In the ancient times Statistics was used to collect information relating to population wealth, military strength etc. Moreover, with the developments in the statistical techniques during the last few decades, today, Statistics is viewed as a means of sound techniques for their handling and analysis and drawing valid inferences from them.

Statistics is important from the following point of views:

- i) Statistics facilitates several functions apart from summarization.
- ii) Statistical methods give tools of comparison
- iii) Statistical methods help in planning, controlling, etc.
- iv) We can get an idea about the shape, symmetry of the data
- v) Statistical methods give systematic methods of data collection and investigation.
- vi) By using statistical methods we can save time, money and manpower.
- vii) Estimation, prediction is also possible using statistical tools.

1.3 Scope of Statistics

- a) Statistics and planning: Statistics are indispensable into planning in the modern age which is termed as “the age of planning”- may it be in business, economics or government level. The statistical data relating to production, consumption, prices, income and expenditure and so on and the advanced statistical techniques handling such data.
- b) Statistics and economics: Statistical data and techniques of statistical analysis have to be immensely useful involving economical problems. Such as wages, price, time series analysis, demand analysis.
- c) Statistics and business: Statistics is an irreplaceable tool of production control. Business executives are relying more and more on statistical techniques for studying the much and desire of the valued customers.
- d) Statistics and industry: In industry statistics is widely used in quality control. In production engineering to find out whether the product is conforming to the specifications or not. Statistical tools, such as inspection plan, control chart etc.
- e) Statistics and mathematics: Statistics are intimately related; recent advancements in statistical technique are the outcome of wide applications of mathematics. The interaction of mathematics and statistics started in the mid seventeenth century with the development of the theory of probability.
- f) Statistics and modern science: In medical science the statistical tools for collection, presentation and analysis of observed facts relating to causes and incidence of diseases and the result of application of various drugs and medicine are of great importance.
- g) Statistics, psychology and education: In education and psychology statistics has found wide application such as, determining or to determine the reliability and validity of a test, factor analysis etc.

1.4.2 Collection of Data :- Collection of data is a vital work and should be done carefully. Prior to gathering the data we need to choose the objective clearly. There are two types of data collection

- a) Primary data:- Primary data is a sort of information that is gathered by specialists straightforwardly from principle sources through interviews, surveys, experiments, etc. Primary data are normally gathered from the source where the data originally originates from and are viewed as the most ideal sort of data in research. So primary data implies original data. This might be viewed as firsthand data. A classical example of primary data is population census results. Primary data is more reliable than any other type however it is costly and tedious. The data which are originally collected for the first time for the purpose of survey are called Primary data.

Methods of collecting Primary data :-

- i) Direct personal investigation
- ii) Indirect investigation
- iii) Through correspondent
- iv) By mailed questionnaire

b)Secondary data :-Secondary data is the data that has effectively been gathered through primary sources and made readily available for researchers to use for their own research. It is a kind of data that has effectively been gathered previously.

Data taken from the sources like office records, reports etc. which are already collected by some other agency is called secondary data. Secondary data are also called finished data because it is already collected, may be tabulated, classified etc.

1.1.Data Types

Basic Concepts of Statistics

1.1.1 Attribute

Definition : An **attribute** is a characteristic or quality that **cannot be measured numerically** but can only be identified or classified.

Attributes describe **qualities** rather than quantities.

Examples : Gender,Blood group,Religion,Marital Status,Nationality.Eye Colour,Literacy etc.

Student	Blood Group
A	O+
B	A+
C	B+

Blood group is an **attribute** because it is a category, not a measurable value. Attributes are measured using nominal and ordinal scale.

a) Nominal Scale: Nominal Scale consists of two or more named categories into which the objects are classified.

For example: i) Classification of students in various divisions of the same standard.

ii) Classification of workers according to skill.

iii) Classification of students according to gender.

b) Ordinal scale: Ordinal Scale of measurement gives numbers to groups of objects using some quantifiable characteristics. Therefore ordered arrangement of groups is possible in this scale.

For example: i) Groups of students according to grades in examinations , such as fail , second class , first class, distinction etc.

ii) Groups using height such as short, medium, tall

iii) Groups of individuals as dull or intelligent

iv) Groups of individuals according to income such as poor, middle class, rich.

Characteristics of Attributes : Non-numerical,Used for classification,Usually expressed in categories and Cannot be measured with units .

1.1.2. Variable

Definition : A **variable** is a characteristic that **can take different numerical values** for different individuals or objects.

Unlike attributes, variables can be measured.

Examples : Age,Height,Weight.Income,Marks ,Temperature etc.

Student	Height (cm)
A	160
B	170
C	165

Height is a **variable** because it is measured numerically.

Characteristics : Measurable, Values may differ from person to person, Used in Statistical calculations etc.

Types of variable: There are two types of variables

a) Quantitative variable

b) Qualitative variable

a) Quantitative variable: Quantitative variables are measured directly such as age, income, production, marks, height, weight etc. Quantitative variable measured numerically in the form of means, percentage. Such variables are called measurement variables.

b) Qualitative variable: Qualitative variable cannot be measured directly but categorized on the basis of presence or absence of attributes. A qualitative variable is called a categorical variable.

Qualitative variables are classified on the basis of descriptive characteristics or attributes.

Example: literates, illiterates, employed, honest etc.

Difference Between Attribute and Variable

Attribute	Variable
Qualitative characteristic	Quantitative characteristic
Cannot be measured numerically	Can be measured numerically
Expressed as categories	Expressed as numbers
Example: Gender	Example: Height

1.1.3. Discrete Variable and Continuous Variable

A. Discrete Variable

Definition: A **discrete variable** takes **countable values**. It cannot have fractions or decimals between two values.

Usually obtained by **counting**.

Examples: Number of students in a class ,Number of books ,Number of cars ,Number of children in a family

Example:

Family	Number of Children
A	2
B	4
C	1

Fractions like 2.5 children are not possible.

B. Continuous Variable:

Definition : A **continuous variable** can take **any value within a given range**, including decimal values.

Usually obtained by **measurement**.

Example 1: Height, Weight, Temperature, Time and Distance etc.

Example 2

Weight of students: 45.5kg, 56.25kg, 72.8kg etc. Any decimal value is possible

Difference Between Discrete and Continuous Variables :

Discrete Variable	Continuous Variable
Obtained by counting	Obtained by measurement
Countable values	Infinite possible values
No decimals	Decimals possible
Example: Number of students	Example: Height

1.1.4. Qualitative and Quantitative Data :

A. Qualitative Data :

Definition : Qualitative data are described by a non-numerical property. Qualitative data describe items in terms of some categories.

Examples: Gender, Blood Group, Religion, Marital Status, Eye colour etc .

Example:

Student	Gender
A	Male
B	Female
C	Male

This is qualitative data because it consists of categories.

B. Quantitative Data :

Definition : The term quantitative data is used to describe a type of information that can be counted or expressed numerically . This type of data is collected in experiments, manipulated and statistically analysed. Quantitative data are numerical and can be measured or counted.

Examples 1: Age, salary, Weight, Height, Marks etc.

Example : 2

Student	Marks
A	75
B	84
C	92

Marks are numerical, so they are quantitative data.

Types of Quantitative Data

1. Discrete Data : Obtained by counting.

Example: Number of students = 45 , Number of cars = 12 .

2. Continuous Data: Obtained by measurement.

Example: Height = 165.5 cm , Weight = 58.75 kg .

Difference Between Qualitative and Quantitative Data:

Qualitative Data	Quantitative Data
Non-numerical	Numerical
Describe qualities	Describe quantities
Cannot perform arithmetic	Arithmetic operations possible
Example: Gender	Example: Age

1.1.5. Classification of Data

Classification is a process related to categorization, the process in which ideas and objects are recognized, differentiated and understood.

After collection of data, the first step is classification. Classification is the grouping of related facts into classes. Classification of statistical data is comparable to the sorting operation. For example, when students seek admission in a college they submit applications to the office. The application forms contain particulars about their performance in the previous examination, their date of birth, sex, nationality etc. If one is interested in finding out how many first, second and third class students have joined the college, one may look into each and every form

and note whether it relates to a first, second class student etc. The process with the help of which this information in a summary form is obtained is called the classification of data.

Objective of classification

The principal objectives of classifying the data are :

- i) To condense the mass of data in such a manner that similarities and dissimilarities can be readily apprehended. Millions of figures can thus be arranged in a few classes having common features.
- ii) To facilitate comparison.
- iii) To point out the most significant features of the data at a glance.
- iv) To focus on the important information collected.
- v) To enable a statistical treatment of the material collected.

Some Definitions:

i) Raw Data: The unprocessed data in terms of individual observations is called raw data.

ii) Frequency: The number of observations in a class is called frequency.

iii) Class limits: The two numbers assigning the class interval are called as class limits.

iv) Class boundaries: The actual class limit of a class interval is called class boundaries.

Class Limit	Class Boundaries
10-19	9.5-19.5
20-29	19.5-29.5
30-39	29.5-39.5

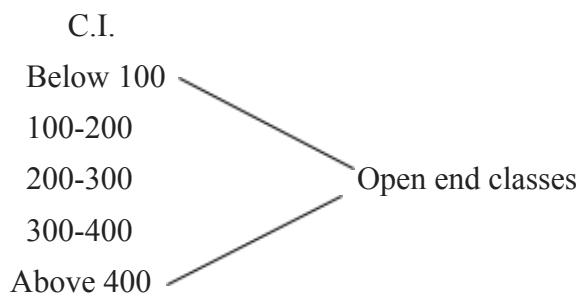
i) Mid values / Class mark: The midpoint of the class interval are called mid values or class mark.i.e. $\text{Mid value} = \frac{\text{upper limit} + \text{lower limit}}{2}$

ii) Class width: The difference between upper limit and lower limit is called class width. Means it is the actual length of class interval.

$$\text{Class width} = \text{Upper limit} - \text{Lower limit}$$

iii) Open end class: A class in which one of the limits is not specified is called an open end class.

For example:



iv) Frequency density:

$$\text{Frequency density} = \frac{\text{Class frequency}}{\text{Class width}}$$

Definition : Classification of data is the process of arranging data into groups or categories according to common characteristics.

Types of Classification

1. Qualitative Classification : Data is classified according to **qualities or attributes**.

Examples : Gender, Religion, Blood group Marital Status etc .

Example :

Student	Blood Group
P	A+
Q	B-
R	B+

2. Quantitative Classification : Data are classified according to **numerical values**.

Example :

Marks	Number of Students
0–20	5
21–40	12
41–60	20
61–80	10
81–100	3

3. Chronological Classification : Data is arranged according to **time**.

Example:

Year	Sales (₹ Lakhs)
2021	25
2022	32
2023	40
2024	48

4. Geographical (Spatial) Classification : Data are arranged according to **location or place**.

Example :

State	Population (Millions)
Maharashtra	124
Gujarat	71
Karnataka	68
Tamil Nadu	76

Advantages of Classification :

- Organizes data systematically.
- Makes data easy to interpret.
- Facilitates comparison.
- Helps in statistical analysis.
- Saves time in decision-making.

1.2.Data Presentation

Data :

The word 'Data 'is frequently used in ordinary conversation. Data is collected in many aspects of everyday life. For example , the time required by a runner to complete a marathon, the

number of shots on goal during a period of hockey , correct and incorrect answers given by a student in an examination etc.

a) Collection of Data: It is the first stage of investigation and is regarding collection of data. It is determined which method of collection is needed in this problem and then data is collected.

b) Organization of Data: It is the second stage. The data are simplified and made comparative and are classified according to time and place.

c) Presentation of Data: In this third stage, organized data is made simple and attractive. These are presented in the form of table diagrams and graphs.

d) Analysis of Data: Fourth stage of investigation is analysis. To get correct results, analysis is necessary. It is often undertaken using Measures of central tendency, Measures of dispersion etc.

e) Interpretation of Data: Interpretation of data is the process of converting analyzed data into meaningful information to draw valid conclusions and make informed decisions.

It makes data easy to understand and analyze.

1.2.1. Frequency Distribution

Definition : The table containing class intervals along with their frequency is called frequency distribution.

Examples :

i)

C.I.	Frequency
0-10	2
10-20	4
20-30	5
30-40	9

ii)

5, 65, 62, 48, 5, 23, 17, 40, 30, 30, 30, 51, 5, 17, 17, 39, 23, 48, 40,
65, 65, 62, 5, 5, 17, 62, 51, 23, 48, 40

Age Group	Tally Marks	Number of people
0 - 10		5
10 - 20		4
20 - 30		3
30 - 40		4
40 - 50		6
50 - 60		2
60 - 70		6
		30

iii)

Scores	Tally marks	Frequency
28		1
31		2
32		1
33		1
41		1
53		5
56		4
61		2
64		3
70		2
74		2
88		1
Total		25

Methods of classification : There are two types of methods classification

i) Inclusive method:- The observation which is equal to the upper limit is included in the same class therefore the method is called an inclusive method.

For Example:

Marks

60-69

70-79

80-89

90-99

Here I observed that the upper limit of one class is not the same as the lower limit of the next class.

ii) Exclusive method :-The observation which is equal to upper limit is not included in the same class which is included in the next class therefore the method is called an exclusive method.

For Example:

Marks

60-70

70-80

80-90

90-100

Here I observed that the upper limit of one class is the same as the lower limit of the next class.

Cumulative Frequencies : A table containing upper limits along with less than type cumulative frequency (l.c.f.) or lower limits along with more than type cumulative frequency (m.c.f) is called cumulative frequency distribution.

C.I.	fi	U.L.	l.c.f.	L.L.	m.c.f
0-20	2	20	2	0	10+28+42+18+2=100

20-40	18	40	2+18=20	20	10+28+42+18=98
40-60	42	60	2+18+42=62	40	10+28+42=80
60-80	28	80	2+18+42+28=90	60	10+28=38
80-100	10	100	2+18+42+28+10=100	80	10
Total	100				

Relative frequency: The proportion of number of observations in a class is the relative frequency
i.e. Relative frequency = $\frac{\text{Class frequency}}{\text{Total frequency}}$

Example:1

Obtain the frequency distribution from the following cumulative frequency distribution

Marks below	10	20	30	40	50
No. of students	1	8	35	46	50

Answer:

C.I.	No. of students(L.C.F.)	Freq.
0-10	1	1
10-20	8	8-1=7
20-30	35	35-8=27
30-40	46	46-35=11
40-50	50	50-46=4

Example:2

Prepare the frequency distribution from the following cumulative frequency distribution

Income more than Rs	500	1000	1500	2000	2500	3000
No. of persons	100	96	92	59	28	2

Answer:

C.I.	No. of persons (M.C.F.)	Freq.
500-1000	100	100-96=4
1000-1500	96	96-92=4
1500-2000	92	92-59=33

2000-2500	59	59-28=31
2500-3000	28	28-2=26
3000-3500	2	2

Example:3

The frequency distribution of daily expenditure of 100 college students is given below

Daily expenditure	50-59	60-69	70-79	80-89	90-99	100-109	110-119	120-129
No. of students	3	10	18	25	24	10	6	4

- State the type of classification.
- Obtain class boundaries of each class.
- Find class width.
- Compute class marks.

Answer:

- Inclusive type of classification.
-

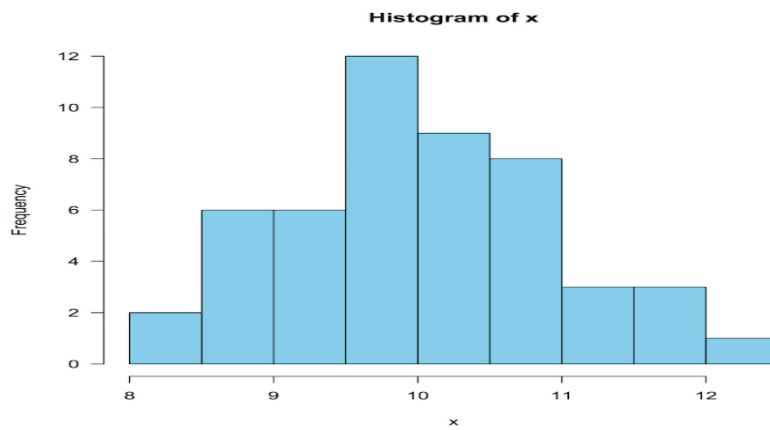
Daily expenditure	Class boundaries	Class width	Class marks
50-59	49.5-59.5	10	54.5
60-69	59.5-69.5	10	64.5
70-79	69.5-79.5	10	74.5
80-89	79.5-89.5	10	84.5
90-99	89.5-99.5	10	94.5
100-109	99.5-109.5	10	104.5
110-119	109.5-119.5	10	114.5
120-129	119.5-129.5	10	124.5

1.2.2. Histogram

Graphs : A frequency distribution can be addressed graphically in any of the accompanying way. Histogram is the most popular and widely used graph. In histogram, grouped data is represented by a series of adjacent rectangles. The base of each rectangle is the class interval. If the classes are not continuous then they are converted into continuous distribution by using class boundaries. Frequencies are taken on Y axis if the class width is the same and if the unequal class width then frequency densities are taken on Y axis.

$$\text{Frequency density} = \frac{\text{Class frequency}}{\text{class width}}$$

We cannot construct histograms for distribution with open end classes.

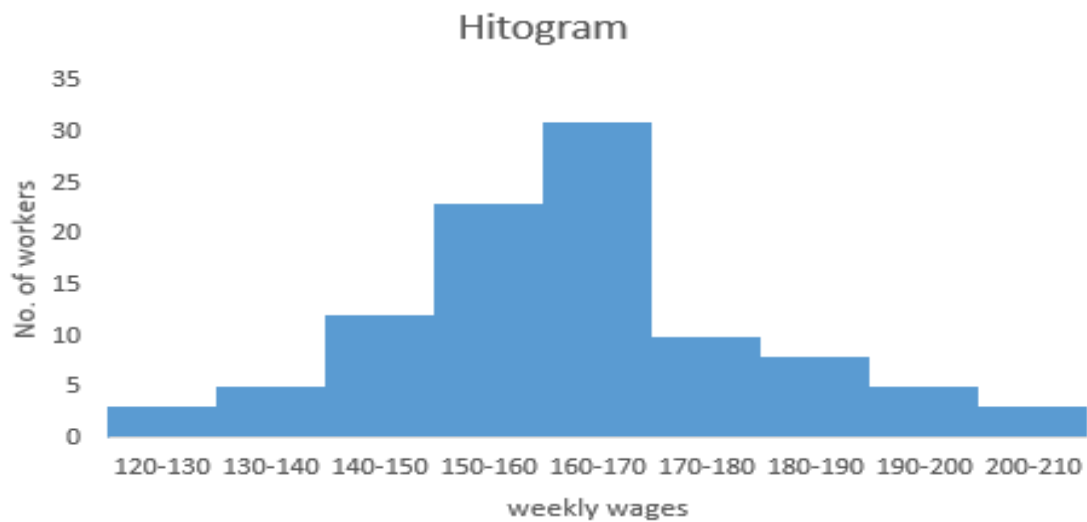


Example:1

Draw the histogram for the following frequency distribution of weekly wages for 100 workers in a factory:

Weekly wages	120-130	130-140	140-150	150-160	160-170	170-180	180-190	190-200	200-210
No. of workers	3	5	12	23	31	10	8	5	3

Answer:

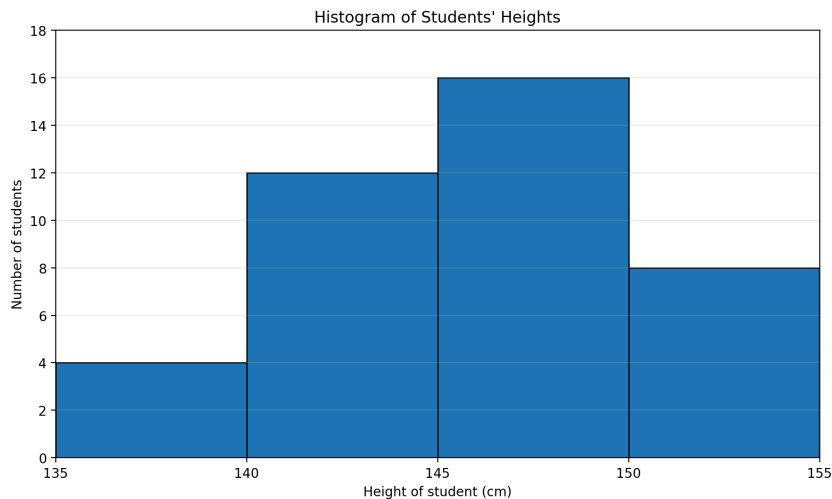


Example : 2

Draw a histogram of the following data.

Height of student (cm)	135-140	140-145	145-150	150-155
No. of students	4	12	16	8

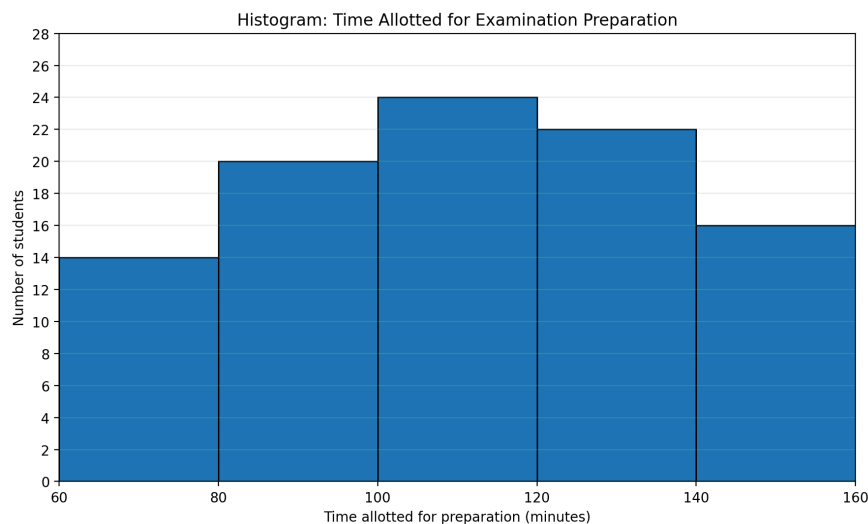
Solution :

**Example : 3**

Time allotted for the preparation of an examination by some students is shown in the table. Draw a histogram to show the information.

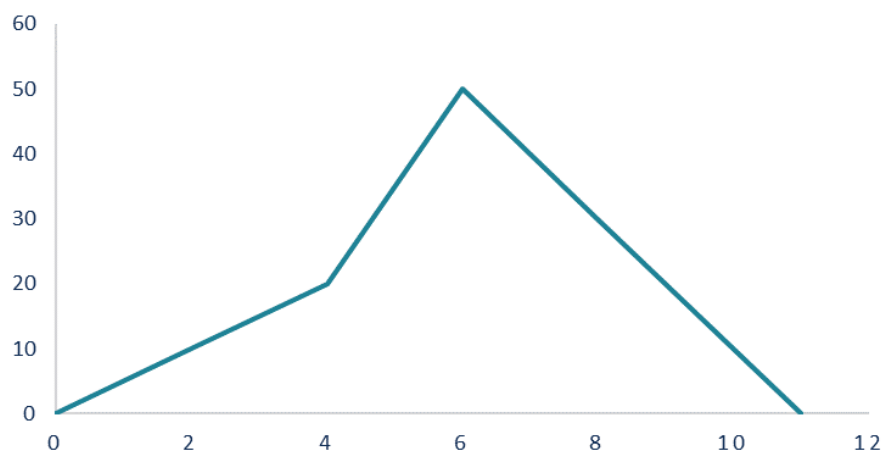
Time (minutes)	60-80	80-100	100-120	120-140	140-160
No. of students	14	20	24	22	16

Solution :

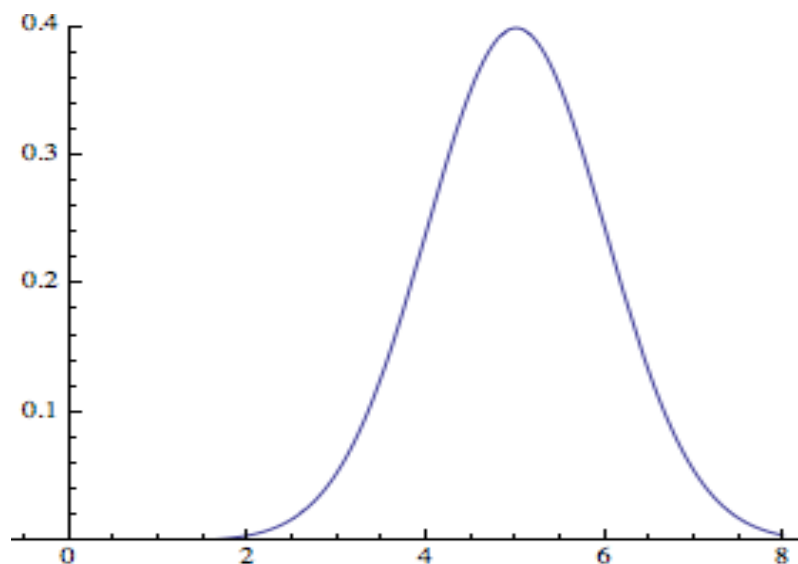


Frequency Polygon:-A frequency polygon is a graph of frequency distribution. In this graph mid values are taken on the X axis and frequencies on the Y axis. Successive points are joined by the line segments. We get a closed figure of the frequency polygon by taking two additional classes, one going before the first class and the other close to the last class with frequency zero. The figure is called frequency polygon.

Remark:-We can draw frequency polygons using histogram. Here mid points of the upper horizontal side of each rectangle join by line segment.



Frequency Curve: There is little contrast in frequency polygon and frequency curve. If the points are joined by smooth curves rather than of straight lines, we get a closed figure called the frequency curve.



Example:1

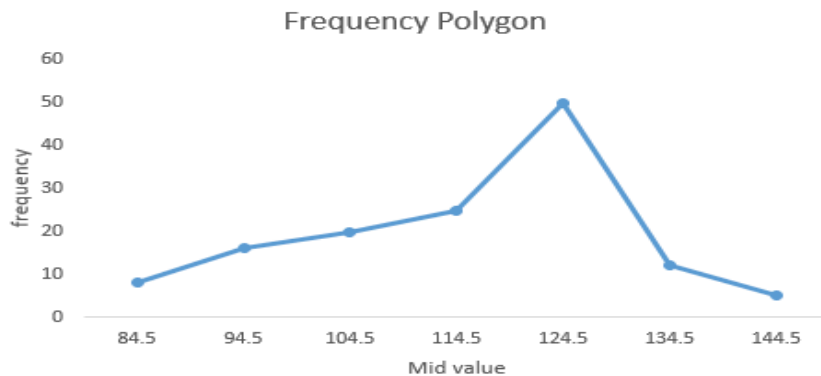
Draw frequency polygon for the following frequency distribution

C.I	80-89	90-99	100-109	110-119	120-129	130-139	140-149
Frequency	8	16	20	25	50	12	5

Answer:

C.I.	Class boundaries	frequency	Mid value
80-89	79.5-89.5	8	84.5
90-99	89.5-99.5	16	94.5
100-109	99.5-109.5	20	104.5
110-119	109.5-119.5	25	114.5
120-129	119.5-129.5	50	124.5

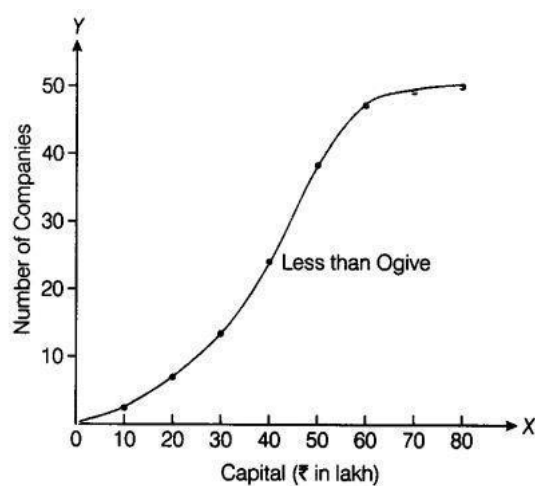
130-139	129.5-139.5	12	134.5
140-149	139.5-149.5	5	144.5



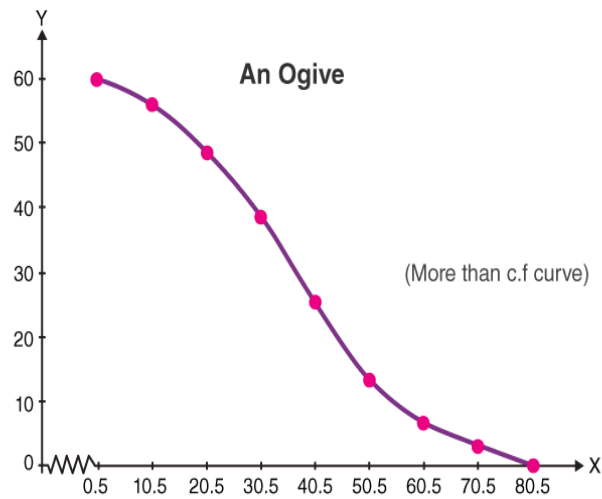
1.2.3.Ogive curve or Cumulative frequency curve:-

Cumulative frequency distribution is represented by cumulative frequency curve. There are two types of cumulative frequencies hence there are two types of cumulative frequency curves.

a) Less than type cumulative frequency curve:- In this case upper limits are taken on X axis and less than type cumulative frequency (l.c.f.) on Y axis. Here we add one more class before the first class with frequency zero for drawing this curve.



b) More than cumulative frequency curve:-In this case lower limits are taken on X axis and more than type cumulative frequency (m.c.f.) on Y axis. Here we add one more class succeeding the last class with frequency zero for drawing this curve.By using this ogive curve we have to find the median.



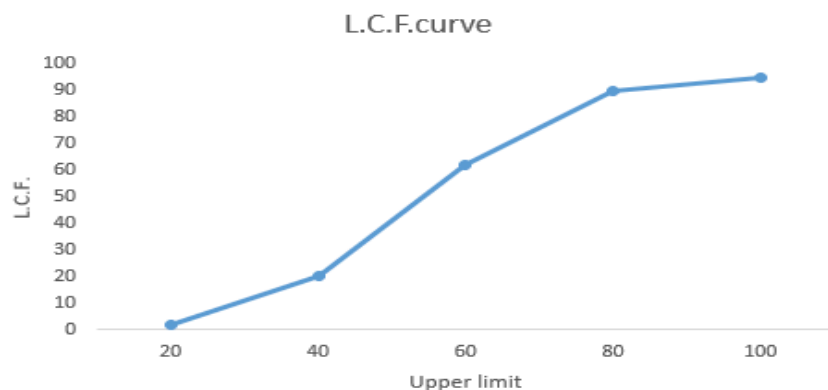
Example:1

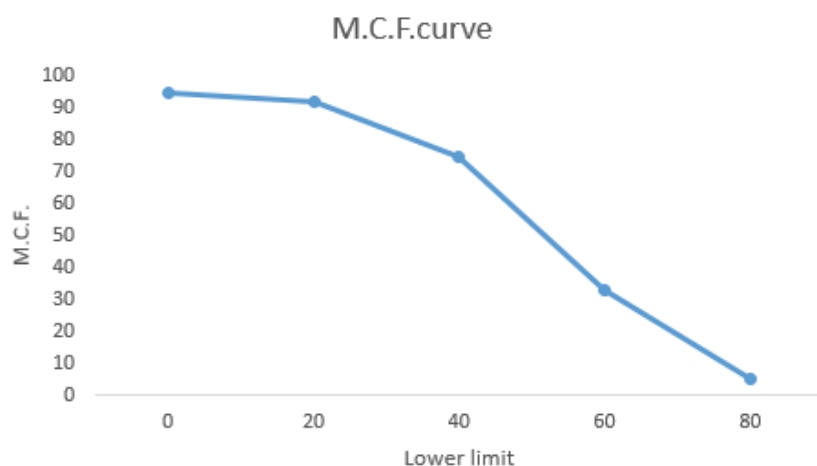
Draw less than cumulative frequency curve and more than cumulative curve for the following frequency distribution

C.I.	0-20	20-40	40-60	60-80	80-100
Frequency	2	18	42	28	5

Answer:-

C.I.	Frequency	U.L.	L.C.F.	L,L	M.C.F.
0-20	2	20	2	0	95
20-40	18	40	20	20	92
40-60	42	60	62	40	75
60-80	28	80	90	60	33
80-100	5	100	95	80	5
Total	95				





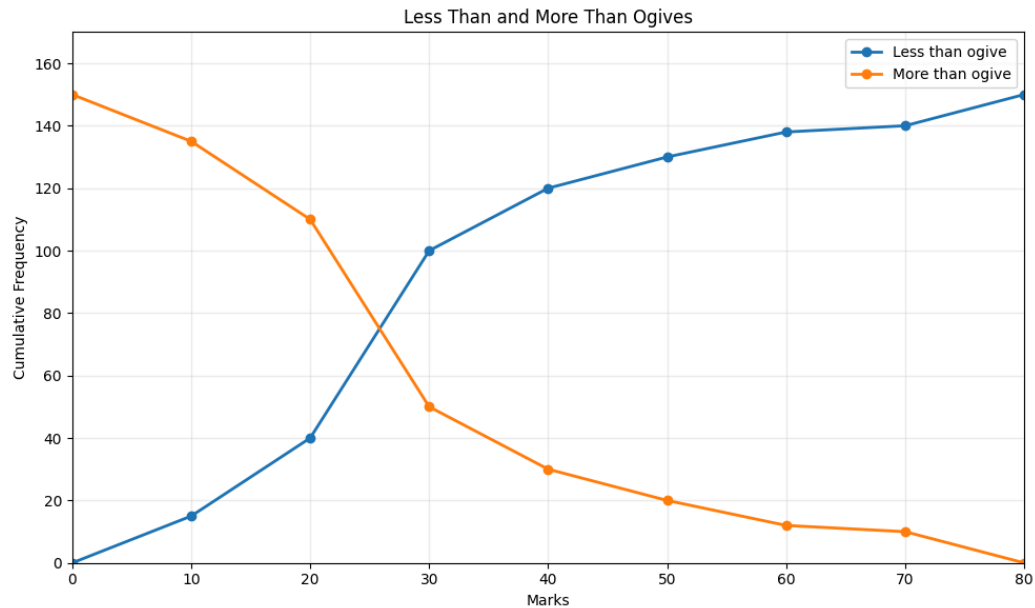
Example : 2

Draw 'Less than ogive' and 'More than ogive' for the following data :

Marks	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No.of Students	15	25	60	20	10	8	2	10

Solution :

Marks	<u>No.of</u> Students (Frequency)	L.T.O	L.T.O C.F.	M.T.O	M.T.O C.F.
0-10	15	10	15	0	150
10-20	25	20	40	10	135
20-30	60	30	100	20	110
30-40	20	40	120	30	50
40-50	10	50	130	40	30
50-60	8	60	138	50	20
60-70	2	70	140	60	12
70-80	10	80	150	70	10
	N=150				



Partition values

Partition values are positional measures that divide a sorted data into equal parts, median, quartiles, deciles, and percentiles are the partition values.

Types of Partition Values

Quartiles (Q): Divide data into 4 equal parts. Q_1 (First/Lower Quartile): Marks the 25th percentile (25% of data lie below). Q_2 (Second Quartile): Marks the 50th percentile (the Median). The 75th percentile (75 percent of data lie below) is represented by Q_3 (Third/Upper Quartile).

Deciles (D): Divide data into 10 equal parts (D_1 through D_9). Each decile segment represents 10% of the dataset.

Percentiles (P): Divide data into 100 equal parts (P_1 through P_{99}). Each percentile represents 1% of the dataset.

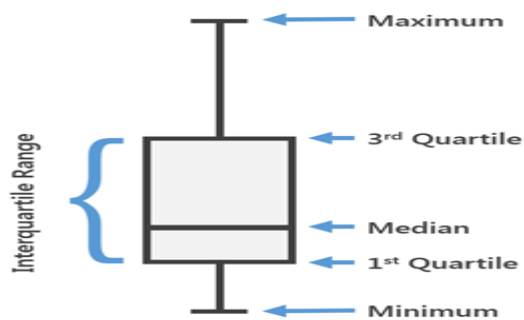
Median: A fundamental partition value that divides the data into two equal halves (50 percent above, 50 percent below), equivalent to Q_2 , D_5 , and P_{50} ,

Example

1.2.4. Box Plot :

The following five numbers are needed to draw a box plot:

1. **Smallest Value:** The minimum number in the data.
1. **First Quartile Q_1 :** About 25% of the data falls below this point.
2. **Median Q_2 :** The exact middle value across the entire data set. 50% of the data is above, and 50% is below.
3. **Third Quartile Q_3 :** About 75% of the data falls below this point.
4. **Largest Value:** The maximum number in the data



Diagrams : It is a technique of introducing numeric data through pictograms, bar diagrams, and pie diagrams. It is the most attractive and engaging approach to address statistical data. Diagrams help in visual comparison and they have a higher perspective.

Example.1 : Make a box -and -whisker plot of the data regarding the length [in sec.] of a famous american song .173,206,179,257,198,251,239,246,295,181,261

Solution : first of all the given data are arranged in ascending order as follows.

173,179,181,198,206,239, 246, 251, 257,261, 295

$$\therefore N=11$$

$$X_{\text{Smallest}} = 173$$

$$X_{\text{largest}} = 295$$

for Q_1

$$Q_1 = \text{value of } \left(\frac{N+1}{4}\right)^{\text{th}} \text{ term}$$

$$= \text{value of } \left(\frac{11+1}{4}\right)^{\text{th}} \text{ term}$$

$$= \text{value of } \left(\frac{12}{4}\right)^{\text{th}} \text{ term}$$

$$Q_1 = \text{value of } 3^{\text{th}} \text{ term}$$

$$\therefore Q_1 = 181$$

for Q_3

$$Q_3 = \text{value of } \left(\frac{3N+1}{4}\right)^{\text{th}} \text{ term}$$

$$= \text{value of } \left(\frac{3(11)+1}{4}\right)^{\text{th}} \text{ term}$$

$$= \text{value of } \left(\frac{34}{4}\right)^{\text{th}} \text{ term}$$

$$Q_3 = \text{value of } (8.5)^{\text{th}} \text{ term}$$

$$Q_3 = \left(\frac{\text{value of (8)th term} + \text{value of 9th term}}{2} \right)$$

$$\therefore Q_3 = 257$$

For Median (M_d)

$$M_d = \text{value of } \left(\frac{N+1}{2}\right)^{th} \text{ term}$$

$$= \text{value of } \left(\frac{11+1}{2}\right)^{th} \text{ term}$$

$$= \text{value of } \left(\frac{12}{2}\right)^{th} \text{ term}$$

$$M_d = \text{value of } 6^{th} \text{ term}$$

$$\therefore M_d = 239$$

Now,

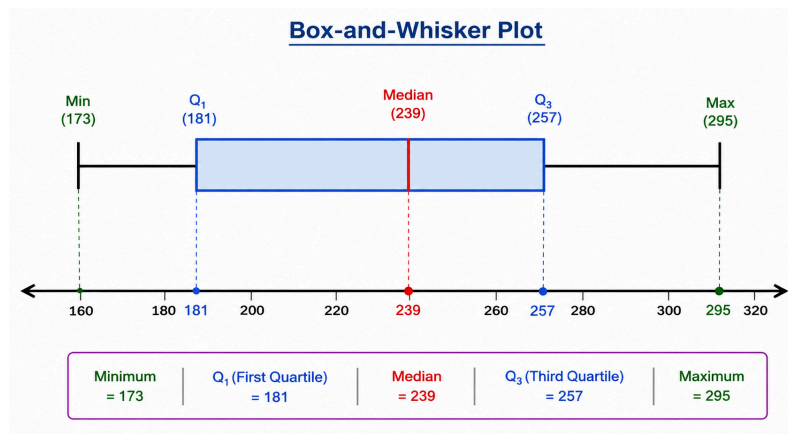
$$X_{\text{Smallest}} = 173$$

$$X_{\text{largest}} = 295$$

$$Q_1 = 181$$

$$Q_3 = 257$$

$$M_d = 239$$



Example 2. Suppose we have a dataset representing the test scores of a group of students: Data (test scores): 78, 85, 90, 92, 95, 96, 97, 98, 99, 100, 105, 110, 120.

Solution : Given data (test scores): 78, 85, 90, 92, 95, 96, 97, 98, 99, 100, 105, 110, 120

First of all, the given data are arranged in ascending order as follows: 78, 85, 90, 92, 95, 96, 97, 98, 99, 100, 105, 110, 120

$$\therefore N = 13$$

$$X_{\text{Smallest}} = 78$$

$$X_{\text{largest}} = 120$$

for Q_1

$$Q_1 = \text{value of } \left(\frac{N+1}{4}\right)^{th} \text{ term}$$

$$= \text{value of } \left(\frac{13+1}{4}\right)^{th} \text{ term}$$

$$= \text{value of } \left(\frac{14}{4}\right)^{th} \text{ term}$$

$Q_1 = \text{value of } 3.5^{th} \text{ term}$

$$Q_1 = \left(\frac{\text{value of } 3^{rd} \text{ term} + \text{value of } 4^{th} \text{ term}}{2} \right)$$

$$Q_1 = \frac{90 + 92}{2}$$

$$\therefore Q_1 = 91$$

for Q_3

$Q_3 = \text{value of } \left(\frac{3N+1}{4} \right)^{th} \text{ term}$

$$= \text{value of } \left(\frac{3(13)+1}{4} \right)^{th} \text{ term}$$

$$= \text{value of } \left(\frac{39+1}{4} \right)^{th} \text{ term}$$

$$Q_3 = \text{value of } (10)^{th} \text{ term}$$

$$\therefore Q_3 = 100$$

For Median (M_d)

$M_d = \text{value of } \left(\frac{N+1}{2} \right)^{th} \text{ term}$

$$= \text{value of } \left(\frac{13+1}{2} \right)^{th} \text{ term}$$

$$= \text{value of } \left(\frac{14}{2} \right)^{th} \text{ term}$$

$M_d = \text{value of } 7 \text{ term}$

$$\therefore M_d = 97$$

Now,

$$X_{\text{Smallest}} = 78$$

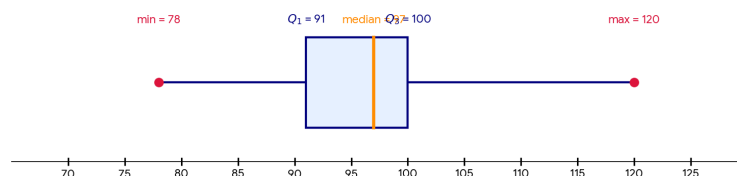
$$X_{\text{largest}} = 120$$

$$Q_1 = 91$$

$$Q_3 = 100$$

$$M_d = 97$$

Proper Box Plot Representation



1.2.5.Bar Plots

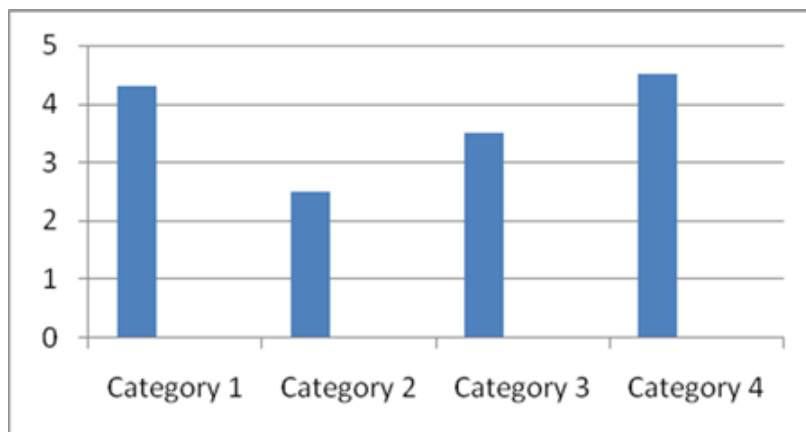
A . Simple Bar Diagram

Practically speaking, a simple bar diagram is extremely famous. A simple bar diagram is used to represent only one variable. For example: Yearly production, Monthly sales etc. In this type of

diagram, year, month etc. are taken on the X-axis and corresponding values of variables are taken on the Y-axis. In this case rectangles of equal width only vary in length.

Remark:

- i) Horizontal bars are used instead of vertical bars. In practice vertical bars are more popular.
- ii) If two or more variables are present the bar diagram cannot be used.

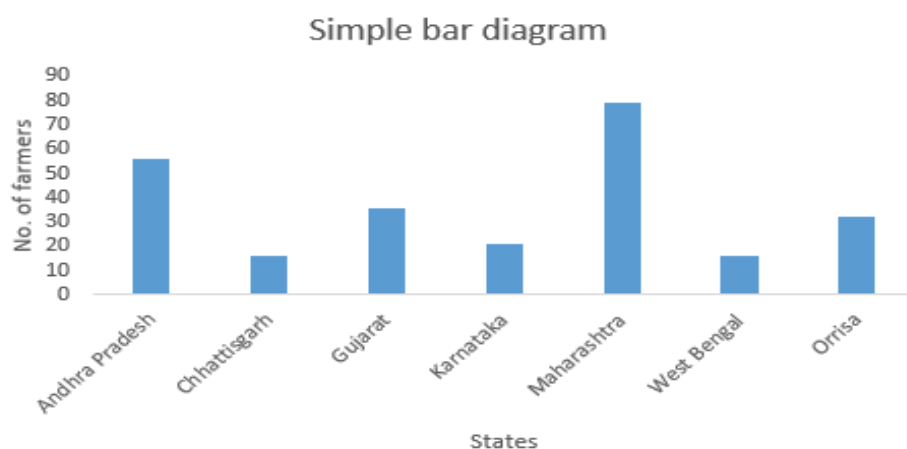


Example:1

The following table shows number of farmers insured between Rabi 1999-2000 and Rabi 2002-03 of different states:

States	No.of farmers insured(lakhs)
Andhra Pradesh	56
Chhattisgarh	16
Gujarat	36
Karnataka	21
Maharashtra	79
West Bengal	16
Orissa	32

Answer:



Example:2

The following table shows the number of marks obtained by Students in a mathematics test.

Students	A	B	C	D	E	F
Marks	50	100	150	200	250	300

Draw a Vertical bar diagram and answer the following question

- Which students scored the highest marks?
- Which students scored the lowest marks?
- What is the difference in marks between students E and Students D?

Example:2

The following table shows the number of marks obtained by Students in a mathematics test.

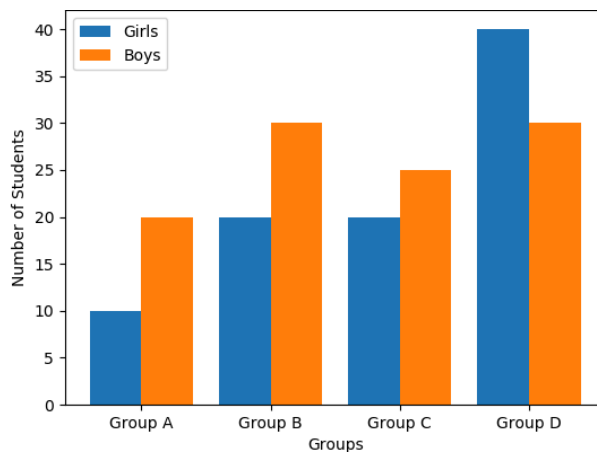
Students	A	B	C	D	E	F
Marks	50	100	150	200	250	300

Draw a Vertical bar diagram and answer the following question

- Which students scored the highest marks?
- Which students scored the lowest marks?
- What is the difference in marks between students E and Students D?

B. Multiple Bar Diagram

Multiple bar diagrams are the same as that of simple bar diagrams. The difference in that since more than one phenomenon is addressed. The different shades, colours are used to distinguish between the bars.

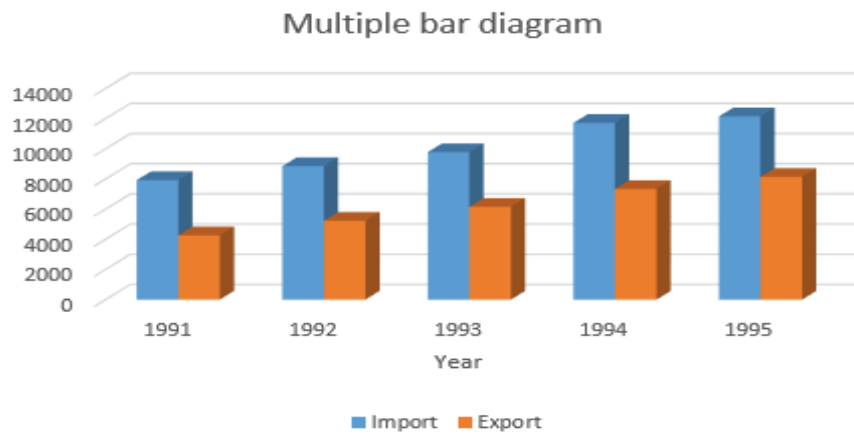


Example:1

Draw multiple bar diagram for the following data:

Year	1991	1992	1993	1994	1995
Import	7930	8850	9780	11720	12150
Export	4260	5225	6150	7340	8145

Answer:



Example:2

The following table shows the birth rate and death rate per 1000 population in a city for different years.

Year	1931-40	1941-50	1951-60	1961-70	1971-80
Birth Rate	46	45	40	42	41
Death Rate	36	31	27	23	19

Draw a multiple bar diagram and answer the question that follow

- in which year was the birth rate highest?
- in which year was the death rate lowest ?
- in which year is the difference between birth rate and death rate maximum ?

Example:3

The following table shows the number of students (in hundreds) enrolled in three different programming courses in a college over five consecutive years.

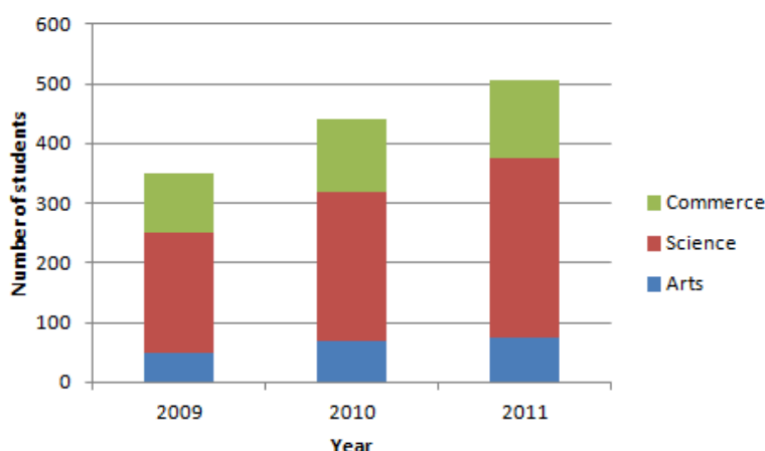
Year	2020	2021	2022	2023	2024
Python	15	18	22	25	28
Java	12	14	16	20	22
Data Science	8	10	12	15	18

Draw a multiple bar diagram and answer the question that follow

- which course has maximum enrollment in the year 2024?
- which course has shown a continuous increase in enrollement ?
- what is the total number of students (in hundreds)enrolled in all courses in 2022?

C. Sub-divided bar diagram:

i) Subdivided bar diagram: If the single variable involves two or more components then we used subdivided bar diagram. Each component occupies a part of the bar proportional to its share in the total. These parts are shown in various colours or shades.
If the number of components are more than 10 or 12 the subdivided bar diagram should not be used.



Example:1

The following table shows the numbers of hydro and thermal power projects (in numbers) in different four year plans of India.

Year	Hydro	Thermal	Total
13th plan (2013-14)	46	64	110
14th plan (2014-15)	49	72	121
15th plan (2015-16)	48	82	130
16th plan (2016-17)	51	89	140

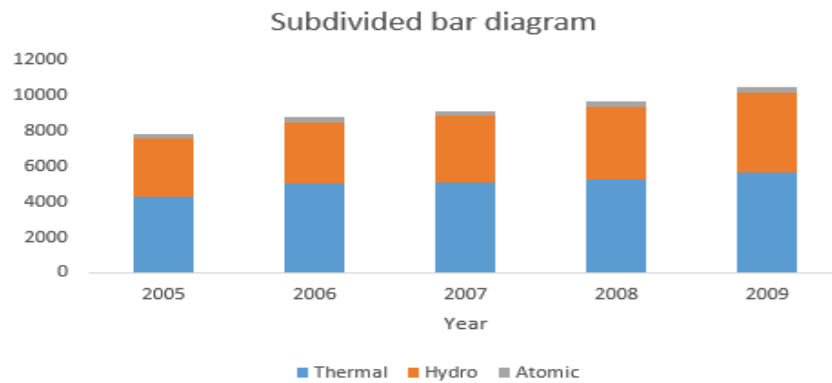
Draw a Sub-divided bar diagram and answer the question that follow

- in which plan was the total number of projects maximum ?
- in which were Thermal projects maximum ?
- Find the percentage of hydro projects in the 15th plan ?

Example:2

Year	2005	2006	2007	2008	2009
Thermal	4300	5000	5100	5300	5700
Hydro	3300	3480	3800	4100	4500
Atomic	260	350	200	300	290

Answer:



1.2.6 .Pi Chart

In this diagram the circle is divided into several sectors. The number of sectors is the same as the number of components.

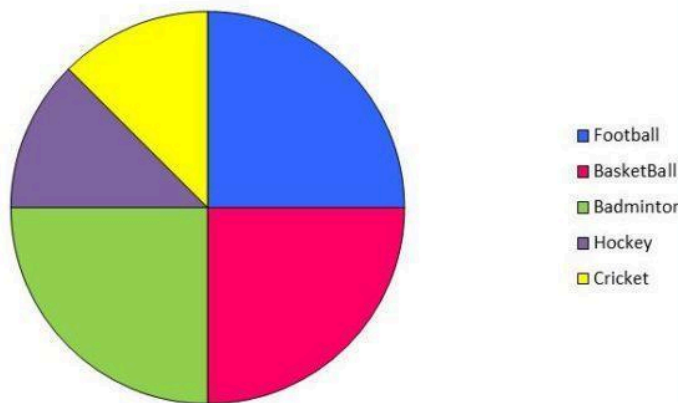
The pie chart is a significant kind of data representation. It contains various segments and areas in which each portion and areas of a pie chart frames a specific segment of the total (percentage). The completeness of the multitude of information is equivalent to 360° . The total value of the pie is consistently 100%.

To work out with the percentage for a pie chart, follow the steps given below:

- i) Sort the information
- ii) Compute the aggregate
- iii) Divide the categories
- iv) Convert into percentage
- v) At long last, compute the degrees

Along these lines, the pie chart formula is given as

$$\frac{\text{Given data}}{\text{Total value of data}} \times 360$$



Example:

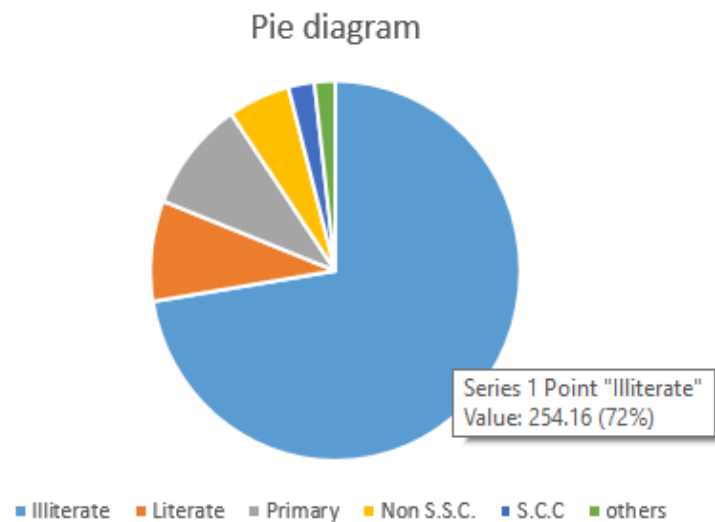
Represent the following data by pie diagram

Education level	Illiterate	Literate	Primary	Non S.S.C.	S.C.C	others
Percentage	70.60	10.85	9.26	5.29	2.23	1.77

Answer:

Education level	Percentage	Angle
-----------------	------------	-------

Illiterate	70.60	254.16
Literate	10.85	30.06
Primary	9.26	33.34
Non S.S.C.	5.29	19.04
S.C.C	2.23	8.03
others	1.77	6.37



Advantages of Graphs

- i) Summarize an enormous dataset in visual structure.
- ii) Easily think about a few informational indexes
- iii) Better explain patterns than do tables.
- iv) Facilitate the comparison.

3.4 Limitations of Graphs

- i) By utilizing graphs we find the values approximately while tables give exact values.
 - ii) Graphs give just a general idea about the phenomenon.
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